Chapter 3

Covalent Bonding in Diatomic Molecules

Although the first two chapters were devoted to presenting the basic principles of quantum mechanics and their application to atomic structure, we must also be concerned with providing information about the structure of molecules. In fact, the structures of molecules constitute the basis for their chemical behavior. The observation that SF_4 reacts very rapidly and vigorously with water whereas SF_6 does not exhibit this behavior is related to the difference in the structures of the molecules. A good understanding of molecular structure is necessary to interpret differences in chemical behavior of inorganic species. Although the chemical formulas for CO_2 and NO_2 may not look much different, the chemistry of these compounds is greatly different. In this chapter, a description of the covalent bond will be presented as it relates to diatomic molecules and their properties. In the next two chapters, bonding in more complex molecules will be described, and the topic of molecular symmetry will be addressed.

3.1 THE BASIC IDEAS OF MOLECULAR ORBITAL METHODS

The formalism that applies to the molecular orbital method will be illustrated by considering a hydrogen molecule, but a more detailed description of both H_2^+ and H_2 will be given in the next section. To begin our description of diatomic molecules, let us imagine that two hydrogen atoms that are separated by a relatively large distance are being brought closer together. As the atoms approach each other, there is an attraction between them that gets greater the shorter the distance between them becomes. Eventually, the atoms reach a distance of separation that represents the most favorable (minimum energy) distance, the bond length in the H_2 molecule (74 pm).

As the distance between the atoms decreases, the nuclei begin to repel each other, as do the two electrons. However, there are forces of attraction between the nucleus in atom 1 and the electron in atom 2 and between the nucleus in atom 2 and the electron in atom 1. We can illustrate the interactions involved as shown in Figure 3.1.

We know that for each atom the ionization potential is 13.6 eV, the bond energy for the H_2 molecule is 4.51 eV (432 kJ mol^{-1}), and the bond length is 74 pm. Keep in mind that bond energies are expressed as the energy necessary to *break* the bond and are therefore *positive* quantities. If the bond *forms*, energy equivalent to the bond energy is released so it is a *negative* quantity.

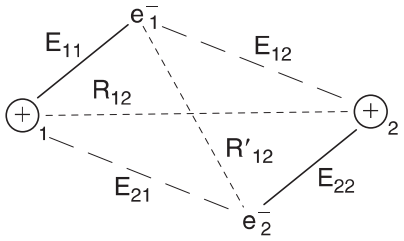


FIGURE 3.1 Interactions within a hydrogen molecule. The quantities represented as R are the repulsions and those as E are the attraction energies. The subscripts indicate the nuclei and electrons involved.

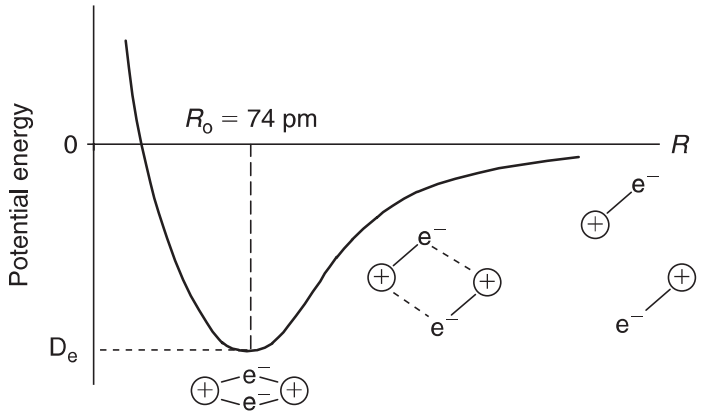


FIGURE 3.2 The interaction of two hydrogen atoms that form a molecule.

The fact that the nuclei do not get closer together does not mean that the forces of attraction and repulsion are equal. The minimum distance is that distance where the total energy (attraction and repulsion) is most favorable. Because the molecule has some vibrational energy, the internuclear distance is not constant, but the equilibrium distance is R_o. Figure 3.2 shows how the energy of interaction between two hydrogen atoms varies with internuclear distance.

In order to describe the hydrogen molecule by quantum mechanical methods, it is necessary to make use of the principles given in Chapter 2. It was shown that a wave function provided the starting point for application of the methods that permitted the calculation of values for the dynamical variables. It is with a *wave function* that we must again begin our treatment of the H_2 molecule by the molecular orbital method. But what wave function do we need? The answer is that we need a wave function for the H_2 *molecule*, and that wave function is constructed from the *atomic* wave functions. The technique used to construct molecular wave functions is known as the linear combination of atomic orbitals (abbreviated as LCAO-MO). The linear combination of atomic orbitals can be written mathematically as

$$\psi = \Sigma\, a_i \phi_i \tag{3.1}$$

In this equation, ψ is the molecular wave function, ϕ is an atomic wave function, and a is a weighting coefficient that gives the relative weight in the "mix" of the atomic wave functions. The summation is

over i, the number of atomic wave functions (number of atoms) being combined. If a diatomic molecule is being described, there are only two atoms involved, so the equation becomes

$$\psi = a_1\phi_1 + a_2\phi_2 \tag{3.2}$$

Although the combination has been written as a sum, the difference is also an acceptable linear combination. The weighting coefficients are variables that must be determined.

In Chapter 2, it was shown that in order to calculate the average value for a dynamical variable a whose operator is α, it is necessary to make use of the relationship

$$\langle a \rangle = \frac{\int \psi^* \alpha \psi \, d\tau}{\int \psi^* \psi \, d\tau} \tag{3.3}$$

If the property we wish to determine is the energy, this equation becomes

$$E = \frac{\int \psi^* \hat{H} \psi \, d\tau}{\int \psi^* \psi \, d\tau} \tag{3.4}$$

where $\hat{H}$ is the Hamiltonian operator, the operator for total energy. The expression shown in Eq. (3.2) is substituted for ψ in the equation above, to give

$$E = \frac{\int (a_1\phi_1^* + a_2\phi_2^*)\hat{H}(a_1\phi_1 + a_2\phi_2)\, d\tau}{\int (a_1\phi_1^* + a_2\phi_2^*)(a_1\phi_1 + a_2\phi_2)\, d\tau} \tag{3.5}$$

When the multiplications are carried out and the constants are removed from the integrals, we obtain

$$E = \frac{a_1^2 \int \phi_1^* \hat{H} \phi_1 \, d\tau + 2a_1a_2 \int \phi_1^* \hat{H} \phi_2 \, d\tau + a_2^2 \int \phi_2^* \hat{H} \phi_2 \, d\tau}{a_1^2 \int \phi_1^* \phi_1 \, d\tau + 2a_1a_2 \int \phi_1^* \phi_2 \, d\tau + a_2^2 \int \phi_2^* \phi_2 \, d\tau} \tag{3.6}$$

In writing this equation, it was assumed that

$$\int \phi_1^* \hat{H} \phi_2 \, d\tau = \int \phi_2^* \hat{H} \phi_1 \, d\tau \tag{3.7}$$

and that

$$\int \phi_1^* \phi_2 \, d\tau = \int \phi_2^* \phi_1 \, d\tau \tag{3.8}$$

These assumptions are valid for a diatomic molecule composed of identical atoms (*homonuclear* diatomic) because ϕ_1 and ϕ_2 are identical and real in this case. In working with the quantities in

equations like Eq. (3.6), certain elements are frequently encountered. For simplicity, the definitions that will be adopted are as follows.

$$H_{11} = \int \phi_1^* \hat{H} \, \phi_1 \, d\tau \tag{3.9}$$

$$H_{12} = \int \phi_1^* \hat{H} \, \phi_2 \, d\tau \tag{3.10}$$

Because $\hat{H}$ is the operator for total energy, H_{11} represents the binding energy of an electron in atom 1 to its nucleus. If the subscripts on the wave functions are both 2, the binding energy of electron 2 to its nucleus is indicated. Such integrals represent the energy of an electrostatic interaction so they are known as *Coulomb integrals*. Integrals of the type shown in Eq. (3.10) indicate the energy of the interaction of the electron in atom 1 with the nucleus in atom 2. Therefore, they are known as *exchange integrals*. As a result of the Hamiltonian being an operator for energy, both types of integrals represent energies. Furthermore, because these integrals represent favorable interactions, they are both negative (representing attractions) in sign.

Because the integral shown in Eq. (3.9) represents the energy with which electron 1 is bound to nucleus 1, it is simply the binding energy of the electron in atom 1. The binding energy of an electron is the reverse (with respect to sign) of the ionization potential. Therefore, it is customary to represent these Coulomb integrals in terms of the ionization potentials by reversing the sign. Although it will not be shown here, the validity of this approximation lies in a principle known as Koopmans' theorem. The valence state ionization potential (VSIP) is generally used to give the value of the Coulomb integral. This assumes that the orbitals are identical in the ion and the neutral atom. However, this relationship is not strictly correct. Suppose an electron is being removed from a carbon atom, which has a configuration of $2p^2$. Because there are two electrons in a set of three orbitals, there are 15 microstates that represent the possible permutations in placing the electrons in the orbitals. There is an exchange energy that is associated with this configuration because of the interchangeability of the electrons in the orbitals, and we say that the electrons are correlated. When one electron is removed, the single electron remaining in the $2p$ orbitals has a different exchange energy, so the measured ionization potential also has associated with it other energy terms related to the difference in exchange energy. Such energies are small compared to the ionization potential, so the VSIP energies are normally used to represent the Coulomb integrals.

The exchange integrals (also knows as resonance integrals) represent the interaction of nucleus 1 with electron 2 and the interaction of nucleus 2 with electron 1. Interactions of this type must be related to the distance separating the nuclei, so the value of an exchange integral can be expressed in terms of the distance separating the atomic nuclei.

In addition to the integrals that represent energies, there are integrals of a type in which no operator occurs. These are represented as

$$S_{11} = \int \phi_1^* \phi_1 \, d\tau \tag{3.11}$$

$$S_{12} = \int \phi_1^* \phi_2 \, d\tau \tag{3.12}$$

Integrals of this type are known as *overlap integrals*, and in a general way, they represent effectiveness with which the orbitals overlap in a region of space. If the subscripts are identical, orbitals on the same atom are indicated, and if the atomic wave functions are normalized, the value of such an integral is 1. As a result, we can write

$$S_{11} = \int \phi_1^* \phi_1 \, d\tau = S_{22} = \int \phi_2^* \phi_2 \, d\tau = 1 \tag{3.13}$$

On the other hand, the integrals of the type

$$S_{12} = \int \phi_1^* \phi_2 \, d\tau = S_{21} = \int \phi_2^* \phi_1 \, d\tau \tag{3.14}$$

are related to the degree of overlap of an orbital on atom 1 with an orbital on atom 2. If the two atoms are separated by a large distance, the overlap integral approaches 0. However, if the atoms are closer together, there is some overlap of the orbitals and $S > 0$. If the atoms were forced together in such a way that the two nuclei coincided (internuclear distance is 0), we would expect $S = 1$ because the orbitals would be congruent. Clearly, the value of an overlap integral like those shown in Eq. (3.14) must be somewhere between 0 and 1, and it must be a function of the internuclear distance. With the exchange integrals and overlap integrals both being functions of the internuclear distance, it should be possible to express one in terms of the other. We will return to this point later.

The appearance of Eq. (3.6) can be simplified greatly by using the notation described earlier. When the substitutions are made, the result is

$$E = \frac{a_1^2 H_{11} + 2a_1 a_2 H_{12} + a_2^2 H_{22}}{a_1^2 + 2a_1 a_2 S_{12} + a_2^2} \tag{3.15}$$

in which we have assumed that S_{11} and S_{22} are both equal to 1 because of the atomic wave functions being normalized. We now seek to find values of the weighting coefficients that make the energy a minimum. To find a minimum in the energy expression, we take the partial derivatives with respect to a_1 and a_2 and set them equal to 0:

$$\left(\frac{\partial E}{\partial a_1}\right)_{a_2} = 0, \left(\frac{\partial E}{\partial a_2}\right)_{a_1} = 0 \tag{3.16}$$

When the differentiations are carried out with respect to a_1 and a_2 in turn while keeping the other constant, we obtain two equations that after simplification can be written as

$$a_1(H_{11} - E) + a_2(H_{12} - S_{12}E) = 0 \tag{3.17}$$

$$a_1(H_{21} - S_{21}E) + a_2(H_{22} - E) = 0 \tag{3.18}$$

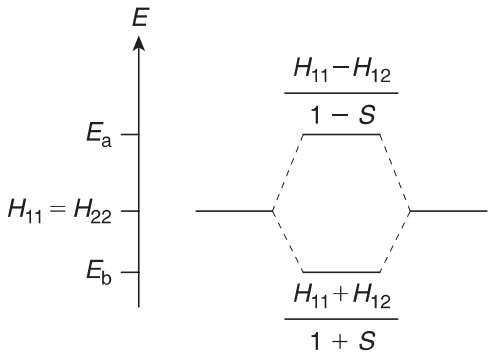


FIGURE 3.3 Combination of two *s* orbitals to produce bonding and antibonding orbitals.

These equations are known as the *secular equations*, and in them the weighting coefficients a_1 and a_2 are the unknowns. These equations constitute a pair of linear equations that can be written in the form

$$ax + by = 0 \quad \text{and} \quad cx + dy = 0 \tag{3.19}$$

It can be shown that a nontrivial solution for a pair of linear equations requires that the determinant of the coefficients must be equal to 0. This means that

$$\begin{vmatrix} H_{11} - E & H_{12} - S_{12}E \\ H_{21} - S_{21}E & H_{22} - E \end{vmatrix} = 0 \tag{3.20}$$

The molecule being described is a homonuclear diatomic, so $H_{12} = H_{21}$ and $S_{12} = S_{21}$. If we represent S_{12} and S_{21} by S and let $H_{11} = H_{22}$, the expansion of the determinant yields

$$(H_{11} - E)^2 - (H_{12} - SE)^2 = 0 \tag{3.21}$$

By equating the two terms on the left-hand side of Eq. (3.21), taking the square root gives

$$H_{11} - E = \pm (H_{12} - SE) \tag{3.22}$$

from which we find two values for E (denoted as E_b and E_a):

$$E_b = \frac{H_{11} + H_{12}}{1 + S} \quad \text{and} \quad E_a = \frac{H_{11} - H_{12}}{1 - S} \tag{3.23}$$

The energy state labeled E_b is known as the *bonding* or *symmetric* state, whereas that designated as E_a is called the *antibonding* or *asymmetric* state. Because both H_{11} and H_{12} are *negative* (binding) energies, E_b represents the state of lower energy. Figure 3.3 shows the qualitative energy diagram for the bonding and antibonding molecular orbitals relative to the 1*s* atomic orbital.

Figure 3.4 shows a more correctly scaled energy level diagram that results for the hydrogen molecule. Note that the energy for the 1*s* atomic orbital of a hydrogen atom is at $-1312\,\text{kJ mol}^{-1}$ because the

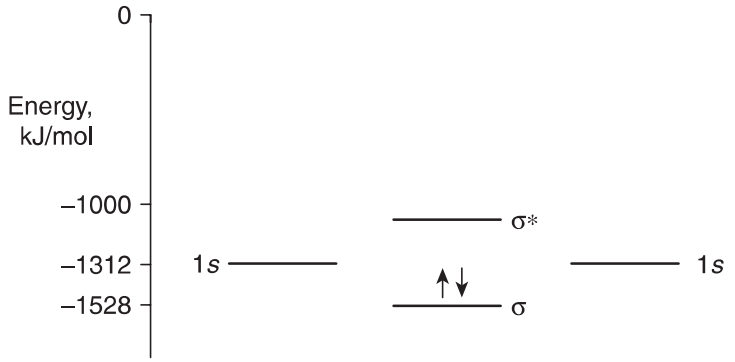


FIGURE 3.4 The energy level diagram for the H_2 molecule.

ionization potential is 1312 kJ mol^{-1} (13.6 eV). Note also that the *bonding* molecular orbital has an energy of −1528 kJ mol^{-1}, which is *lower* than that of the 1*s* state.

If a hydrogen molecule is separated into the two constituent atoms, the result is equivalent to taking the two electrons in the bonding molecular orbital and placing them back in the atomic orbitals. Because there are two electrons, that energy would be $2(1528 - 1312) = 432$ kJ mol^{-1}, the bond energy of the H_2 molecule. From the molecular orbital diagram, it can be seen that although the energy of the antibonding state is higher than that of the hydrogen atom, it is still very *negative*. An energy of 0 does not result because even when the atoms are completely separated, the energy of the system is the sum of the binding energies in the atoms, which is $2(-1312)$ kJ mol^{-1}. The bonding and antibonding states are "split" above and below the energy state of the electron in an atom, not above and below an energy of 0. However, the antibonding state is raised a greater amount than the bonding state is lowered relative to the atomic orbital energy. This can be seen from the relationships shown in Eqs. (3.23) because in the first case the denominator is $(1 + S)$ while in the other it is $(1 - S)$.

If we substitute the values for the energy as shown in Eq. (3.23) into the secular equations, we find that

$$a_1 = a_2 \text{(for the } \textit{symmetric} \text{ state) and } a_1 = -a_2 \text{ (for the } \textit{antisymmetric} \text{ state)}$$

When these relationships between the weighting coefficients are used, it is found that

$$\psi_b = a_1\phi_1 + a_2\phi_2 = \frac{1}{\sqrt{2 + 2S}}(\phi_1 + \phi_2) \tag{3.24}$$

$$\psi_a = a_1\phi_1 - a_2\phi_2 = \frac{1}{\sqrt{2 - 2S}}(\phi_1 - \phi_2) \tag{3.25}$$

If we let A represent the normalization constant, the condition for normalization is that

$$1 = \int A^2(\phi_1 + \phi_2)\,d\tau = A^2\left[\int \phi_1^2\,d\tau + \int \phi_2^2\,d\tau + 2\int \phi_1\phi_2\,d\tau\right] \tag{3.26}$$

The first and second integrals on the right-hand side of this equation evaluate to 1 because the atomic wave functions are assumed to be normalized. Therefore, the right-hand side of the equation reduces to

$$1 = A^2[1 + 1 + 2S] \tag{3.27}$$

and we find that the normalization constant is given by

$$A = \frac{1}{\sqrt{2 + 2S}} \tag{3.28}$$

and the wave functions can be represented as shown in Eqs. (3.24) and (3.25).

Although we have dealt with a diatomic molecule consisting of two hydrogen atoms, the procedure is exactly the same if the molecule is Li_2, except that the atomic wave functions are 2*s* wave functions and the energies involved are those appropriate to lithium atoms. The VSIP for lithium is only 513 kJ mol^{-1} rather than 1312 kJ mol^{-1} as it is for hydrogen.

When performing molecular orbital calculations, overlap and exchange integrals must be evaluated. With modern computing techniques, overlap integrals are most often evaluated as part of the calculation. The wave functions are of the Slater type (see Section 2.4), and the overlap integrals can be evaluated for varying bond lengths and angles. Many years ago, it was common for the values of overlap integrals to be looked up in a massive set of tables that presented the values of the overlap integrals for the various combinations of atomic orbitals and internuclear distances. These tables, known as the Mulliken tables, were prepared by R. A. Mulliken and coworkers, and they were essential for giving part of the data needed to perform molecular orbital calculations.

The exchange integrals, H_{ij}, are evaluated by representing them as functions of the Coulomb integrals, H_{ii}, and the overlap integrals. One such approximation is known as the Wolfsberg-Helmholtz approximation, which is written as

$$H_{12} = -KS\left(\frac{H_{11} + H_{22}}{1 + S}\right) \tag{3.29}$$

where H_{11} and H_{22} are the Coulomb integrals for the two atoms, S is the overlap integral, and K is a constant having a numerical value of approximately 1.75. Because the overlap integral is a function of the bond length, so is the exchange integral.

The quantity $(H_{11} + H_{22})$ may not be the best way to combine Coulomb integrals for cases where the atoms have greatly different ionization potentials. In such cases, it is preferable to use the Ballhausen-Gray approximation,

$$H_{12} = -KS(H_{11}H_{22})^{\frac{1}{2}} \tag{3.30}$$

Another useful approximation for the H_{12} integral is that known as the Cusachs approximation and is written as

$$H_{12} = \frac{1}{2} S(K - |S|)(H_{11} + H_{22}) \tag{3.31}$$

Although the energy of a chemical bond as a function of internuclear distance can be represented by the potential energy curve shown in Figure 3.2, neither the Wolfsberg-Helmholtz nor the Ballhausen-Gray approximation is a function that possesses a minimum. However, the approximation of Cusachs is a mathematical expression that does pass through a minimum.

It has already been shown that the energy of the bonding molecular orbital can be written as $E_b = (H_{11} + H_{12})/(1 + S)$. Suppose there are two electrons in a bonding orbital having this energy and the bond is being broken. If the bond breaking is homolytic (one electron ends up on each atom), the electrons will reside in atomic orbitals having an energy of H_{11} (which is the same as H_{22} if a homonuclear molecule is being disrupted). Before the bond is broken, the two electrons have a total energy that is given by the expression $2[(H_{11} + H_{12})/(1 + S)]$, and after the bond is broken the binding energy for the two electrons is $2H_{11}$. Therefore, the bond energy (BE) can be expressed as

$$BE = 2H_{11} - 2\left(\frac{H_{11} + H_{12}}{1 + S}\right) \tag{3.32}$$

In order to use this equation to calculate a bond energy, it is necessary to have the values for H_{12} (the values for the H_{ii} integrals are usually available by approximating from the ionization potentials) and S. To a rough approximation (known as neglecting the overlap), the value of S can be assumed to be 0 because the value is small (in the range 0.1 to 0.4) in many cases.

3.2 THE H_2^+ AND H_2 MOLECULES

The simplest diatomic molecule consists of two nuclei and a single electron. That species, H_2^+, has properties some of which are well known. For example, in H_2^+ the internuclear distance is 104 pm and the bond energy is 268 kJ/mol. Proceeding as illustrated in the previous section, the wave function for the bonding molecular orbital can be written as

$$\psi_b = a_1\phi_1 + a_2\phi_2 = \frac{1}{\sqrt{2 + 2S}}(\phi_1 + \phi_2) \tag{3.33}$$

This wave function describes a bonding orbital of the σ type that arises from the combination of two 1s wave functions for atoms 1 and 2. To make that point clear, the wave function could be written as

$$\psi_b(\sigma) = a_1\phi_{1(1s)} + a_2\phi_{2(1s)} = \frac{1}{\sqrt{2 + 2S}}\left(\phi_{1(1s)} + \phi_{2(1s)}\right) \tag{3.34}$$

The foregoing expression is actually a one-electron wave function, which is adequate in this case but not for the H_2 molecule. The energy associated with this molecular orbital can be calculated as shown in Eq. (3.4), from which we obtain

$$E[\psi_b(\sigma)] = \int [\psi_b(\sigma)]\ \hat{H}\ [\psi_b(\sigma)]\ d\tau \tag{3.35}$$

When the approximation is made that the overlap can be neglected, $S = 0$ and the normalization constant is $1/2^{1/2}$. Therefore, after substituting the results shown in Eq. (3.34) for $\psi_b(\sigma)$ the expression for the energy of the molecular orbital can be written as

$$E[\psi_b(\sigma)] = \frac{1}{2}\int (\phi_{1(1s)} + \phi_{2(1s)})\ \ \hat{H}(\phi_{1(1s)} + \phi_{2(1s)})\ d\tau \tag{3.36}$$

Separating the integral gives

$$E[\psi_b(\sigma)] = \frac{1}{2}\int \phi_{1(1s)}\ \hat{H}\phi_{1(1s)}\ d\tau + \frac{1}{2}\int \phi_{2(1s)}\ \hat{H}\ \phi_{2(1s)}\ d\tau + \frac{1}{2}\int \phi_{1(1s)}\ \hat{H}\,\phi_{2(1s)}\ d\tau + \frac{1}{2}\int \phi_{2(1s)}\ \hat{H}\,\phi_{1(1s)}\ d\tau \tag{3.37}$$

As we saw earlier, the first two terms on the right-hand side of this equation represent the electron binding energies in atoms 1 and 2, respectively, which are H_{11} and H_{22}, the Coulomb integrals. The last two terms represent the exchange integrals, H_{12} and H_{21}. In this case, $H_{11} = H_{22}$ and $H_{12} = H_{21}$ because the nuclei are identical. Therefore, the energy of the orbital is

$$E[\psi_b(\sigma)] = \tfrac{1}{2}H_{11} + \tfrac{1}{2}H_{11} + \tfrac{1}{2}H_{12} + \tfrac{1}{2}H_{12} = H_{11} + H_{12} \tag{3.38}$$

Although the development will not be shown here, the energy of the antibonding orbital can be written as

$$E[\psi_a(\sigma)] = \tfrac{1}{2}H_{11} + \tfrac{1}{2}H_{11} - \tfrac{1}{2}H_{12} - \tfrac{1}{2}H_{12} = H_{11} - H_{12} \tag{3.39}$$

We have already noted that in the case where S = 0, the molecular orbitals reside above and below the atomic states by an amount H_{12}. Using this approach, the calculated values for the bond energy and internuclear distance in H_2^+ do not agree well with the experimental values. Improved values are obtained when an adjustment in the total positive charge of the molecule is made as was done in the case of the helium atom. For H_2^+, it turns out that the electron is acted on as if the total nuclear charge were about 1.24 rather than 2. This is essentially the same approach as that taken in the case of the helium atom (see Section 2.3). Also, the molecular orbital wave function was constructed by taking a linear combination of 1*s* atomic wave functions. A better approach is to take an atomic wave function that contains not only *s* character but also a contribution from the $2p_z$ orbital that lies along the internuclear axis. When these changes are made, the agreement between the calculated and experimental properties of H_2^+ is much better.

The wave functions just described are one-electron wave functions, but the H_2 molecule has two electrons to be dealt with. In the methods of molecular orbital theory, a wave function for the

two electrons in the hydrogen molecule is formed by taking the product of the two one-electron wave functions. Therefore, the wave function for the bonding molecular orbital in H_2 is written in terms of the atomic wave functions, ϕ, as

$$\psi_{b,1}\psi_{b,2} = [\phi_{A,1} + \phi_{B,1}][\phi_{A,2} + \phi_{B,2}] \tag{3.40}$$

In this case, the subscript b indicates the bonding (σ) orbital, A and B subscripts denote the two nuclei, and subscripts 1 and 2 denote electrons 1 and 2, respectively. Expanding the expression on the right-hand side of Eq. (3.40) gives

$$\psi_{b,1}\psi_{b,2} = \phi_{A,1}\ \phi_{B,2} + \phi_{A,2}\ \phi_{B,1} + \phi_{A,1}\ \phi_{A,2} + \phi_{B,1}\ \phi_{B,2} \tag{3.41}$$

In this expression, the term $\phi_{A,1}\ \phi_{B,2}$ essentially represents the interaction of two 1*s* orbitals for hydrogen atoms A and B. The term $\phi_{A,2}\ \phi_{B,1}$ represents the same type of interaction with the electrons interchanged. However, the term $\phi_{A,1}\ \phi_{A,2}$ represents *both* electrons 1 and 2 interacting with nucleus A. That means the structure described by the wave function is ionic, $H_A^-\ H_B^+$. In an analogous way, the term $\phi_{B,1}\ \phi_{B,2}$ represents both electrons interacting with nucleus B, which corresponds to the structure $H_A^+\ H_B^-$. Therefore, what we have devised for a molecular wave function actually describes the hydrogen molecule as a "hybrid" (a valence bond term that is applied incorrectly) of

$$H_A : H_B \leftrightarrow H_A{:}^-\ H_B^+ \leftrightarrow H_A^+ : H_B^-$$

As in the case of the H_2^+ molecule, the calculated properties (bond energy and bond length) of the H_2 molecule do not agree well with the experimental values when this wave function is utilized. An improvement is made by allowing the nuclear charge to be a variable with the optimum value being about 1.20. Additional improvement between calculated and experimental values is achieved when the atomic wave functions are not considered to be pure 1*s* orbitals but rather allow some mixing of the 2*p* orbitals. Also, the wave function shown in Eq. (3.41) makes no distinction in the weighting given to the covalent and ionic structures. Our experience tells us that for identical atoms that have the same electronegativity, an ionic structure would not be nearly as significant as a covalent one. Therefore, weighting parameters should be introduced that adjust the contributions of the two types of structures to reflect the chemical nature of the molecule.

The preceding discussion is presented in order to show how the basic ideas of the molecular orbital approach are employed. It is also intended to show how to approach getting improved results after the basic ideas are used to generate molecular wave functions. For the purposes here, it is sufficient to indicate the nature of the changes rather than presenting quantitative results of the calculations.

A simple interpretation of the nature of a covalent bond can be seen by considering some simple adaptations of the wave function. For example, it is ψ^2 that is related to probability of finding the electrons. When we write the wave function for a bonding molecular orbital as ψ_b, that means that because $\psi_b = \phi_A + \phi_B$,

$$\psi_b^2 = (\phi_A + \phi_B)^2 = \phi_A^2 + \phi_B^2 + 2\ \phi_A\phi_B \tag{3.42}$$

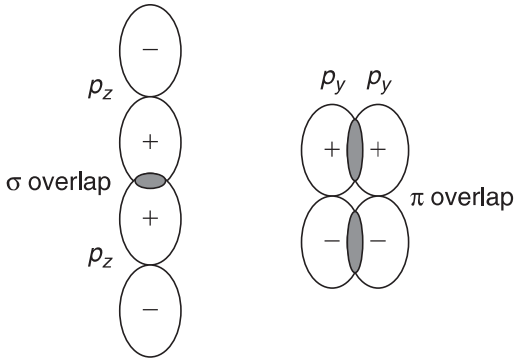


FIGURE 3.5 Possible ways for overlap of *p* orbitals to occur.

Because the last term (when integration is performed over all space) becomes

$$\int \phi_A \phi_B \, d\tau$$

which is actually an overlap integral. The expression in Eq. (3.42) indicates that there is an *increased* probability of finding the electrons between the two nuclei as a result of orbital overlap. This is, of course, for the *bonding* molecular orbital. For the antibonding orbital the combination of atomic wave functions is written as

$$\psi_a^2 = (\phi_A - \phi_B)^2 = \phi_A^2 + \phi_B^2 - 2\,\phi_A \phi_B \tag{3.43}$$

This expression indicates that there is a decreased probability (indicated by the term $-2\phi_A\phi_B$) of finding the electrons in the region between the two nuclei. In fact, there is a nodal plane between the positive and negative (with respect to algebraic sign) of the two regions of the molecular orbital. As a simple definition, we can describe a covalent bond as the increased probability of finding electrons between two nuclei or an increase in electron density between the two nuclei when compared to the probability or density that would exist simply because of the presence of two atoms.

3.3 DIATOMIC MOLECULES OF SECOND-ROW ELEMENTS

The basic principles dealing with the molecular orbital description of the bonding in diatomic molecules have been presented in the previous section. However, somewhat different considerations are involved when second-row elements are involved in the bonding because of the differences between *s* and *p* orbitals. When the orbitals being combined are *p* orbitals, the lobes can combine in such a way that the overlap is symmetric around the internuclear axis. Overlap in this way gives rise to a σ bond. This type of overlap involves *p* orbitals for which the overlap is essentially "end on" as shown in Figure 3.5. For reasons that will become clear later, it will be assumed that the p_z orbital is the one used in this type of combination.

The essential idea is that orbital lobes of the same sign can lead to favorable overlap (the overlap integral has a value >0). This can occur between orbitals of different types in several ways. Figure 3.6 shows a few of the types of orbital overlap that lead to bonding. As we shall see in later chapters, some of these types are quite important.

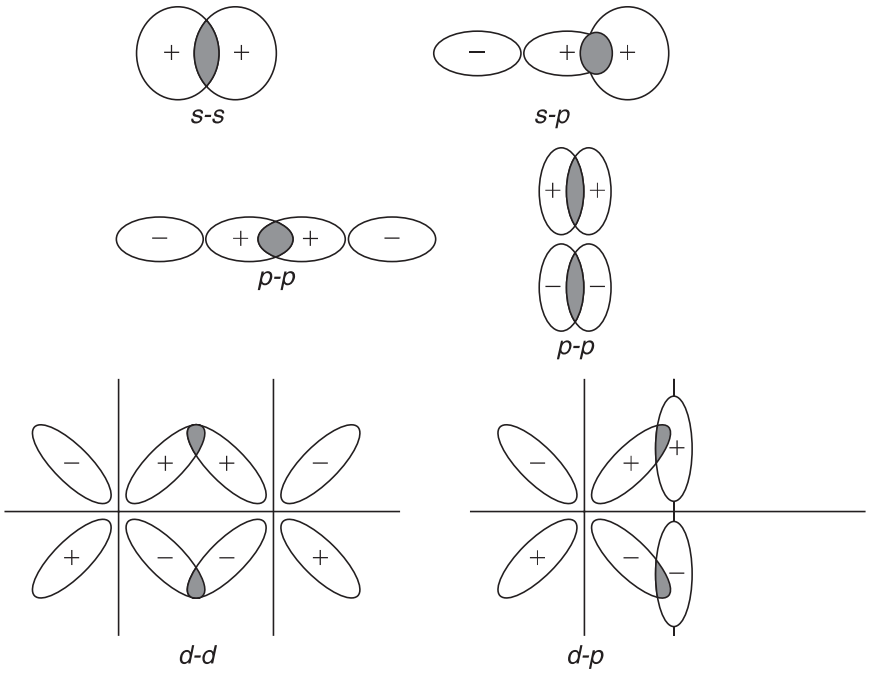


FIGURE 3.6 Some types of orbital overlap that lead to energetically favorable interactions.

Representing the p_z orbitals on atoms 1 and 2 by z_1 and z_2, the combinations of atomic wave functions can be shown as

$$\psi(\sigma_z) = \frac{1}{\sqrt{2 + 2S}}\left[\phi(z_1) + \phi(z_2)\right] \tag{3.44}$$

$$\psi(\sigma_z)^* = \frac{1}{\sqrt{2 - 2S}}\left[\phi(z_1) - \phi(z_2)\right] \tag{3.45}$$

After the σ bond has formed, further interaction of the p orbitals on the two atoms is restricted to the p_x and p_y orbitals, which are perpendicular to the p_z orbital. When these orbitals interact, the region of orbital overlap is not symmetrical around the internuclear axis but rather on either side of the internuclear axis, and a π bond results. Orbital overlap of this type is also shown in Figures 3.5 and 3.6. The combinations of wave functions for the bonding π orbitals can be written as

$$\psi(\pi_x) = \frac{1}{\sqrt{2 + 2S}}\left[\phi(x_1) + \phi(x_2)\right] \tag{3.46}$$

$$\psi(\pi_y) = \frac{1}{\sqrt{2 + 2S}}\left[\phi(y_1) + \phi(y_2)\right] \tag{3.47}$$

The two bonding π orbitals represented by these wave functions are degenerate. The wave functions for the antibonding states are identical in form except that negative signs are used in the combination of atomic wave functions and in the normalization constants.

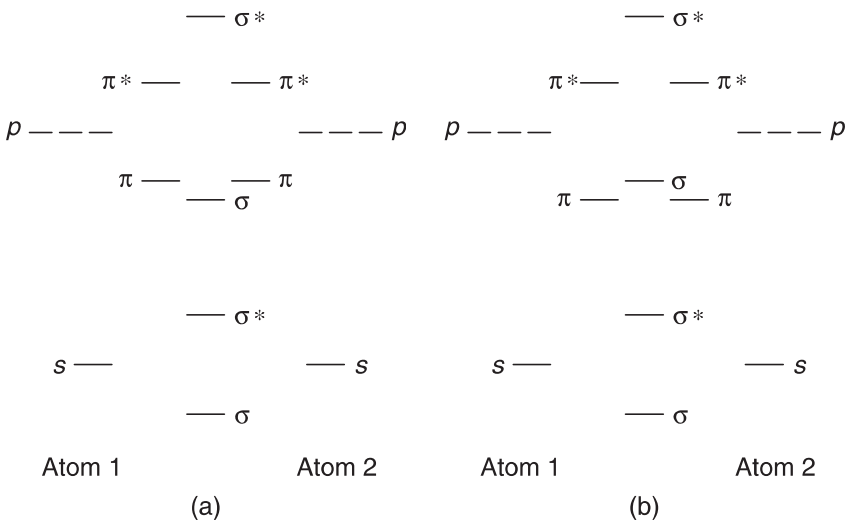


FIGURE 3.7 Energy level diagrams for diatomic molecules of second-row elements. Early members of the series follow the diagram shown in (b), whereas later members follow (a).

The combination of three p orbitals on one atom with three p orbitals on another leads to the formation of one σ and two π bonding molecular orbitals. The order in which the orbitals are populated is

$$\sigma(2s)\ \sigma^*(2s)\ \sigma(2p_z)\ \pi(2p_x)\ \pi(2p_y)\ \ldots\ .$$

It might be assumed that the $\sigma(2p_z)$ orbital would always have a lower energy than the two π orbitals, but that is not necessarily the case. Orbitals of similar energy interact best when combining as a result of hybridization. Mixing of the $2s$ and $2p_z$ orbitals is allowed in terms of symmetry (see Chapter 5), but the $2s$ and $2p_x$ or $2p_y$ results in zero overlap because they are orthogonal. For the elements early in the second period where the nuclear charge is low, the $2s$ and $2p$ orbitals are similar in energy, so it is possible for them to hybridize extensively. For the later members of the group (N, O, and F), the higher nuclear charge causes the difference in energy between the $2s$ and $2p$ orbitals to be great enough that they cannot hybridize effectively. One result of hybridizing orbitals is that their energies are changed, and in this case that results in the order of filling the σ and π orbitals to be reversed so that for B_2 and C_2, there is experimental evidence to show that the π orbitals lie lower in energy than the σ orbital. For the atoms later in the second period, the extent of hybridization of the $2s$ and $2p$ orbitals is slight, which results in the σ orbital lying lower in energy than the two π orbitals. The order of filling the molecular orbitals is

$$\sigma(2s)\ \sigma^*(2s)\ \pi(2p_x)\ \pi(2p_y)\ \sigma(2p_z)\ \ldots$$

Figure 3.7 shows both of the molecular orbital energy diagrams that result for diatomic molecules of second-row elements.

The fact that the B_2 molecule is paramagnetic shows that the highest occupied molecular orbitals (usually abbreviated as HOMO) are the degenerate π orbitals, each of which is occupied by one electron.

Further evidence that this is the correct energy level scheme to be used for C_2 comes from the fact that the molecule is diamagnetic. The molecular orbital configurations for these molecules can be written as

$$\begin{aligned} B_2 &\quad (\sigma)^2\ (\sigma^*)^2\ (\pi)^1\ (\pi)^1 \\ C_2 &\quad (\sigma)^2\ (\sigma^*)^2\ (\pi)^2\ (\pi)^2 \end{aligned}$$

Sometimes, these configurations are shown with the atomic orbitals indicated from which the molecular orbitals arise. For example, B_2 could be described as

$$B_2 \quad [\sigma(2s)]^2\ [(\sigma^*(2s)]^2\ [\pi(2p_x)]^1\ [\pi(2p_y)]^1$$

Although the subject of symmetry has not yet been discussed in this book, it is true that σ orbitals that are bonding in character have "*g*" symmetry because the wave functions are symmetric with respect to the center of the bond. Essentially, this means that if $\psi(x, y, z)$ is equal to $\psi(-x, -y, -z)$, the function is said to be an even function or to have *even parity*. This is signified by "g," which comes from the German word *gerade*, meaning "even." If $\psi(x, y, z)$ is equal to $-\psi(-x, -y, -z)$ the function has *odd parity* and it is indicated by "u" which comes from the word *ungerade*, meaning "uneven." An atomic *s* orbital is g whereas a *p* orbital is u in symmetry. Although a bonding σ orbital is g, π bonding orbitals have u symmetry because they are antisymmetric with respect to the internuclear axis. Antibonding molecular orbitals of each type have the symmetry labels reversed. Sometimes the symmetry character of the molecular orbital is indicated by means of a subscript. When this is done, the representation for B_2 is

$$B_2 \quad (\sigma_g)^2\ (\sigma_u)^2\ (\pi_u)^1\ (\pi_u)^1$$

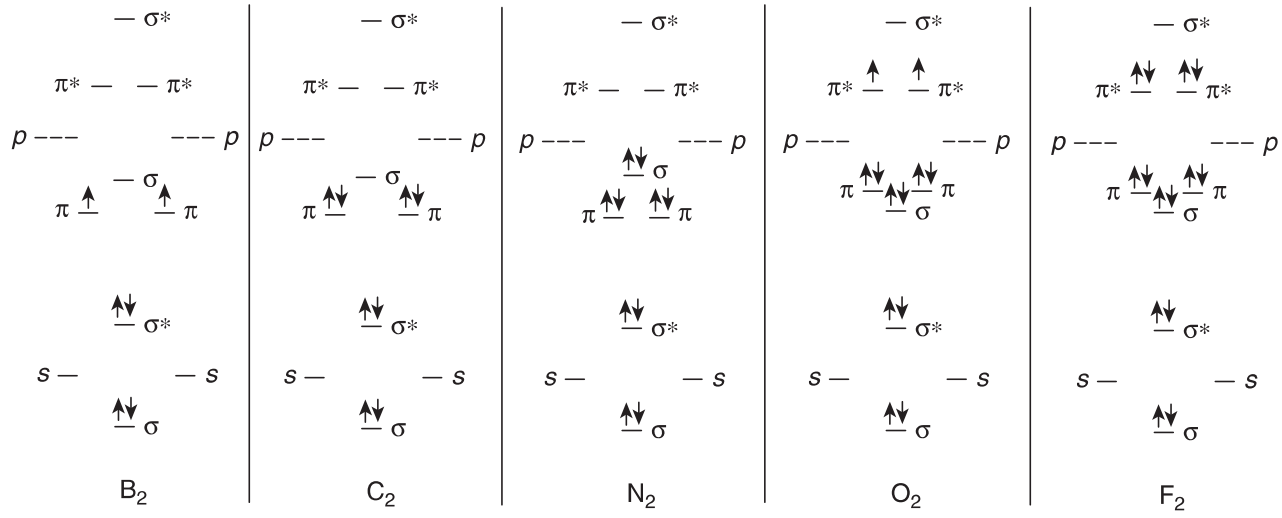
Molecular orbitals are sometimes given numerical prefixes to show the order in which the orbitals having those type and symmetry designations are encountered. When this is done, the order of filling the molecular orbitals for the second-row elements is shown as

$$1\sigma_g\ 1\sigma_u\ 2\sigma_g\ 1\pi_u\ 1\pi_u\ 1\pi_g\ 1\pi_g\ 2\sigma_u$$

In this case, "1" indicates the first instance where an orbital of that type is encountered. A "2" indicates the second time an orbital having the designation following the number is encountered. The various ways to identify molecular orbitals are shown here because different schemes are sometimes followed by different authors.

In writing these configurations for diatomic molecules of second-row elements, we have omitted the electrons from the 1*s* orbitals because they are not part of the valence shells of the atoms. When considering the oxygen molecule, for which the σ orbital arising from the combinations of the $2p_z$ orbitals lies lower in energy than the π orbitals, we find that the electron configuration is

$$O_2 \quad (\sigma)^2\ (\sigma^*)^2\ (\sigma)^2\ (\pi)^2\ (\pi)^2\ (\pi^*)^1\ (\pi^*)^1$$



	B_2	C_2	N_2	O_2	F_2
B.O.	1	2	3	2	1
R, pm	159	131	109	121	142
B.E., eV	3.0	5.9	9.8	5.1	1.6

FIGURE 3.8 Molecular orbital diagrams for second-row homonuclear diatomic molecules.

Note that there are two unpaired electrons in the degenerate π^* orbitals, and as a result, the oxygen molecule is paramagnetic. Figure 3.8 shows the orbital energy diagrams for the diatomic molecules of the second-row elements. Keep in mind that in these molecular orbital diagrams the atomic orbitals are not all at the same energy, so neither do the molecular orbitals of the same type have the same energy for different molecules.

A concept that is important when considering bonds between atoms is the *bond order*, B. The bond order is a measure of the *net* number of electron pairs used in bonding. It is related to the number of electrons in bonding orbitals (N_b) and the number in antibonding orbitals (N_a) by the equation

$$B = \frac{1}{2}(N_b - N_a) \tag{3.48}$$

The bond order for each diatomic molecule is given in Figure 3.8, as is the bond energy. Note that there is a general increase in bond energy as the bond order increases. This fact makes it possible to see why certain species behave as they do. For example, the bond order for the O_2 molecule is $(8 - 4)/2 = 2$, and we say that a bond order of 2 is equivalent to a double bond. If an electron is removed from an oxygen molecule, the species O_2^+ results, and the electron removed comes from the highest occupied orbital, which is a π^* (antibonding) orbital. The bond order for O_2^+ is $(8 - 3)/2 = 2.5$, which is higher than that of the O_2 molecule. Because of this, it is not at all unreasonable to expect there to be some reactions in which an oxygen molecule reacts by losing an electron to form O_2^+, the dioxygenyl cation. Of course, such a reaction would require the reaction of oxygen with a very strong oxidizing

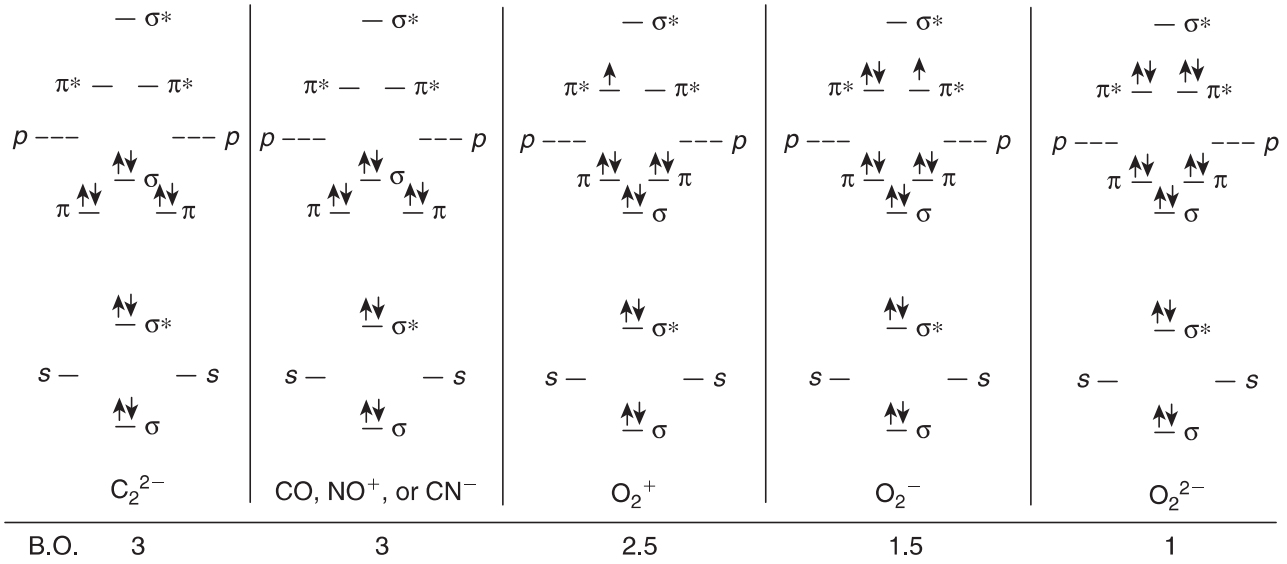


FIGURE 3.9 Molecular orbital diagrams for some heteronuclear molecules and ions of second-row elements.

agent. One such oxidizing agent is PtF_6 which contains platinum in the +6 oxidation state. The reaction with oxygen can be written as

$$PtF_6 + O_2 \rightarrow O_2^+ + PtF_6^- \tag{3.49}$$

Although this reaction shows the formation of O_2^+, it is also possible to add one electron to the O_2 molecule to produce O_2^-, the superoxide ion, or two electrons to form O_2^{2-}, the peroxide ion. In each case, the electrons are added to the antibonding π^* orbitals, which reduces the bond order from the value of 2 in the O_2 molecule. For O_2^- the bond order is 1.5, and it is only 1 for O_2^{2-}, the peroxide ion. The O–O bond energy in the peroxide ion has a strength of only 142 kJ mol^{-1} and, as expected, most peroxides are very reactive compounds. The superoxide ion is produced by the reaction

$$K + O_2 \rightarrow KO_2 \tag{3.50}$$

In addition to the homonuclear molecules, the elements of the second period form numerous important and interesting heteronuclear species, both neutral molecules and diatomic ions. The molecular orbital diagrams for several of these species are shown in Figure 3.9. Keep in mind that the energies of the molecular orbitals having the same designations are not equal for these species. The diagrams are only qualitatively correct.

It is interesting to note that both CO and CN^- are isoelectronic with the N_2 molecule. That is, they have the same number and arrangement of electrons as the N_2 molecule. However, as we will see later, these species are quite different from N_2 in their chemical behavior. The properties of many homonuclear and heteronuclear molecules and ions are presented in Table 3.1.

Table 3.1 Characteristics of Some Diatomic Species.

Species	N_b	N_a	B^a	R, pm	DE[b], eV
H_2^+	1	0	0.5	106	2.65
H_2	2	0	1	74	4.75
He_2^+	2	1	0.5	108	3.1
Li_2	2	0	1	262	1.03
B_2	4	2	1	159	3.0
C_2	6	2	2	131	5.9
N_2	8	2	3	109	9.76
O_2	8	4	2	121	5.08
F_2	8	6	1	142	1.6
Na_2	2	0	1	308	0.75
Rb_2	2	0	1	—	0.49
S_2	8	4	2	189	4.37
Se_2	8	4	2	217	3.37
Te_2	8	4	2	256	2.70
N_2^+	7	2	2.5	112	8.67
O_2^+	8	3	2.5	112	6.46
BN	6	2	2	128	4.0
BO	7	2	2.5	120	8.0
CN	7	2	2.5	118	8.15
CO	8	2	3	113	11.1
NO	8	3	2.5	115	7.02
NO^+	8	2	3	106	—
SO	8	4	2	149	5.16
PN	8	2	3	149	5.98
SiO	8	2	3	151	8.02
LiH	2	0	1	160	2.5
NaH	2	0	1	189	2.0
PO	8	3	2.5	145	5.42

[a] *B is the bond order,* $(N_a - N_b)/2$.
[b] *DE is the dissociation energy (1 eV = 96.48 kJ mol*$^{-1}$*).*

3.4 PHOTOELECTRON SPECTROSCOPY

Most of what we know about the structure of atoms and molecules has been obtained by studying the interaction of electromagnetic radiation with matter. Line spectra reveal the existence of shells of different energy where electrons are held in atoms. From the study of molecules by means of infrared spectroscopy, we obtain information about vibrational and rotational states of molecules. The types of bonds present, the geometry of the molecule, and even bond lengths may be determined in specific cases. The spectroscopic technique known as photoelectron spectroscopy (PES) has been of enormous importance in determining how electrons are bound in molecules. This technique provides direct information on the energies of molecular orbitals in molecules.

In PES, high-energy photons are directed to the target from which electrons are ejected. The photon source that is frequently employed is the He(I) source, which emits photons having an energy of 21.22 eV as the excited state $2s^1\ 2p^1$ relaxes to the $1s^2$ ground state. The ionization potential for the hydrogen atom is 13.6 eV, and the first ionization potential for many molecules is of comparable magnitude. The principle on which PES works is that a photon striking an electron causes the electron to be ejected. The kinetic energy of the ejected electron will be

$$\frac{1}{2}mv^2 = h\nu - I \tag{3.51}$$

where $h\nu$ is the energy of the incident photon and I is the ionization potential for the electron. This situation is somewhat analogous to the photoelectric effect (see Section 1.2). In PES, a molecule, M, is ionized by a photon,

$$h\nu + \mathrm{M} \rightarrow \mathrm{M}^+ + \mathrm{e}^- \tag{3.52}$$

Electrons that are ejected are passed through an analyzer, and by means of a variable voltage, electrons having different energies can be detected. The numbers of electrons having specific energies are counted, and a spectrum showing the number of electrons emitted (intensity) versus energy is produced. In most cases, when an electron is removed during ionization, most molecules are in their lowest vibrational state. Spectra for diatomic molecules show a series of closely spaced peaks that correspond to ionization that leads to ions that are in excited vibrational states. If ionization takes place with the molecule in its lowest vibrational state to produce the ion in its lowest vibrational state, the transition is known as an *adiabatic ionization*. When a diatomic molecule is ionized, the most intense absorption corresponds to ionization with the molecule and the resulting ion having the same bond length (see Section 17.6). This is known as the *vertical ionization*, and it leads to the ion being produced in excited vibrational states. In general, the molecule and the ion have nearly identical bond lengths when the electron is ejected from a nonbonding orbital.

Applications of the PES technique to molecules have yielded an enormous amount of information regarding molecular orbital energy levels. For example, PES has shown that the bonding π orbitals in oxygen are higher in energy than the σ orbital arising from the combination of the $2p$ wave functions. For nitrogen, the reverse order of orbitals is found. When electrons are ejected from the bonding σ_{2p} orbital of O_2, two absorption bands are observed. There are two electrons populating that orbital, one

with a spin of +½ and the other with a spin of −½. If the electron removed has a spin of −½, the electron having a spin of +½ remains, and it can interact with the two electrons in the π^* orbitals that have spins of +½. This can be shown as follows, where $(\sigma)^{1(+½)}$ means that there is one electron having a spin of +½ in the σ orbital, and so forth.

$$O_2^+ \quad (\sigma)^2\ (\sigma^*)^2\ (\sigma)^{1(+½)}\ (\pi)^2\ (\pi)^2\ (\pi^*)^{1(1+½)}\ (\pi^*)^{1(+½)}$$

If the electron removed from the σ orbital has a spin of +½, the resulting O_2^+ ion is

$$O_2^+ \quad (\sigma)^2\ (\sigma^*)^2\ (\sigma)^{1(-½)}\ (\pi)^2\ (\pi)^2\ (\pi^*)^{1(+½)}\ (\pi^*)^{1(+½)}$$

These two O_2^+ ions have slightly different energies, as is exhibited by their photoelectron spectra. Studies such as these have contributed greatly to our understanding of molecular orbital energy diagrams. We will not describe the technique further, but more complete details of the method and its use can be found in the references at the end of this chapter.

3.5 HETERONUCLEAR DIATOMIC MOLECULES

Atoms do not all have the same ability to attract electrons. When two different types of atoms form a covalent bond by sharing a pair of electrons, the shared pair of electrons will spend more time in the vicinity of the atom that has the greater ability to attract them. In other words, the electron pair is *shared*, but it is not shared *equally*. The ability of an atom in a molecule to attract electrons to it is expressed as the *electronegativity* of the atom. Earlier, for a homonuclear diatomic molecule we wrote the combination of two atomic wave functions as

$$\psi = a_1\phi_1 + a_2\phi_2 \tag{3.53}$$

where we did not have to take into account the difference in the ability of two atoms to attract electrons. For two different types of atoms, we can write the wave function for the bonding molecular orbital as

$$\psi = \phi_1 + \lambda\phi_2 \tag{3.54}$$

where the parameter λ is a weighting coefficient. Actually, a weighting coefficient for the wave function of one atom is assumed to be 1, and a different weighting factor, λ, is assigned for the other atom depending on its electronegativity.

When two atoms share electrons unequally, it means that the bond between them is polar. Another way to describe this is to say that the bond has *partial ionic character*. For the molecule AB, this is equivalent to drawing two structures, one of which is covalent and the other ionic. However, there are actually three structures that can be drawn

$$\begin{matrix} A{:}B & \leftrightarrow & A^+B^- & \leftrightarrow & A^-B^+ \\ \text{I} & & \text{I} & & \text{III} \end{matrix} \tag{3.55}$$

If we write a wave function for the molecule to show a combination of these structures, it is written as

$$\psi_{\text{molecule}} = a\psi_{\text{I}} + b\psi_{\text{II}} + c\psi_{\text{III}} \tag{3.56}$$

where a, b, and c are constants and ψ_{I}, ψ_{II}, and ψ_{III} are wave functions that correspond to the structures I, II, and III, respectively. Generally, we have some information about the magnitudes of a, b, and c. For example, if the molecule being considered is HF, the resonance structure H^- F^+ will contribute very little to the actual structure of the molecule. It is contrary to the chemical nature of the H and F atoms to have a structure with a negative charge on H and a positive charge on F. Accordingly, the weighting coefficient for structure III must be approximately 0. For molecules that are predominantly covalent in nature, even structure II will make a smaller contribution than will structure I.

The dipole moment, μ, for a diatomic molecule (the situation for polyatomic molecules that have several bonds is more complex) can be expressed as

$$\mu = q \times r \tag{3.57}$$

where q is the quantity of charge separated and r is the distance of separation. If an electron were completely transferred from one atom to the other, the quantity of charge separated would be e, the charge on an electron. For bonds in which an electron pair is shared unequally, q is less than e, and if the sharing is equal there is no charge separation, $q = 0$, and the molecule is nonpolar. For a polar molecule, there is only one bond length, r. Therefore, the ratio of the actual or observed dipole moment (μ_{obs}) to that assuming complete transfer of the electron (μ_{ionic}) will give the ratio of the amount of charge separated to the charge of an electron:

$$\frac{\mu_{\text{obs}}}{\mu_{\text{ionic}}} = \frac{q \cdot r}{e \cdot \text{r}} = \frac{q}{e} \tag{3.58}$$

The ratio q/e gives the fraction of an electron that appears to be transferred from one atom to another. This ratio can also be considered as the partial ionic character of the bond between the atoms. It follows that the percent of ionic character is 100 times the fraction of ionic character. Therefore,

$$\%\ \text{Ionic character} = \frac{100\,\mu_{\text{obs}}}{\mu_{\text{ionic}}} \tag{3.59}$$

The actual structure of HF can be represented as a composite of the covalent structure H–F, in which there is equal sharing of the bonding electron pair, and the ionic structure H^+ F^-, where there is complete transfer of an electron from H to F. Therefore, the wave function for the HF molecule wave function can be written in terms of the wave functions for those structures as

$$\psi_{\text{molecule}} = \psi_{\text{covalent}} + \lambda\psi_{\text{ionic}} \tag{3.60}$$

The squares of the coefficients in a wave function are related to probability. Therefore, the total contribution from the two structures is $1^2 + \lambda^2$ while the contribution from the ionic structure is given by λ^2. As a result, $\lambda^2/(1^2 + \lambda^2)$ gives the fraction of ionic character to the bond and

$$\% \text{ Ionic character} = \frac{100\,\lambda^2}{(1 + \lambda^2)} \tag{3.61}$$

because $1^2 = 1$. Therefore,

$$\frac{\mu_{\text{obs}}}{\mu_{\text{ionic}}} = \frac{\lambda^2}{(1 + \lambda^2)} \tag{3.62}$$

For the HF molecule, the bond length is 0.92 Å (0.92×10^{-8} cm $= 0.92 \times 10^{-10}$ m) and the measured dipole moment is 1.91 debye or 1.91×10^{-18} esu cm. If an electron were completely transferred from H to F, the dipole moment (μ_{ionic}) would be

$$\mu_{\text{ionic}} = 4.80 \times 10^{-10} \text{ esu} \times 0.92 \times 10^{-8} \text{ cm} = 4.41 \times 10^{-18} \text{ esu cm} = 4.41 \text{ D}$$

Therefore, the ratio $\mu_{\text{obs}}/\mu_{\text{ionic}}$ is 0.43, which means that

$$0.43 = \frac{\lambda^2}{(1 + \lambda^2)} \tag{3.63}$$

from which we find that $\lambda = 0.87$. Therefore, the wave function for the HF molecule can be written as

$$\psi_{\text{molecule}} = \psi_{\text{covalent}} + 0.87\,\psi_{\text{ionic}} \tag{3.64}$$

From the foregoing analysis, it appears that we can consider the polar HF molecule as consisting of a hybrid made from a purely covalent structure contributing 57% and an ionic structure contributing 43% to the actual structure:

$$\begin{array}{ccc} \text{H : F} & \leftrightarrow & \text{H}^+\text{F}^- \\ 57\% & & 43\% \end{array}$$

Of course, HF is actually a polar covalent molecule, but from the extent of the polarity, it behaves *as if it were* composed of the two structures shown above. A similar analysis can be carried out for all of the hydrogen halides, and the results are shown in Table 3.2.

A simple interpretation of the effect of two atoms in a diatomic molecule is seen from the molecular orbital description of the bonding. Different atoms have different ionization potentials, which results in the values for the Coulomb integrals used in a molecular orbital calculation being different. In fact, according to Koopmans' theorem, the ionization potential with the sign changed gives the value for the Coulomb integral. In terms of a molecular orbital energy level diagram, the atomic states of the two atoms are different and the bonding molecular orbital will be closer in energy to that of the atom having the higher ionization potential. For example, in the HF molecule, there is a single

Table 3.2 Parameters for Hydrogen Halide Molecules, HX.

Molecule	*r*, pm	μ_{obs}, D	μ_{ionic}, D	% Ionic character (100 μ_{obs}/μ_{ionic})	$\chi_X - \chi_H$
HF	92	1.91	4.41	43	1.9
HCl	128	1.03	6.07	17	0.9
HBr	143	0.78	6.82	11	0.8
HI	162	0.38	7.74	5	0.4

1 debye 10^{-18} esu cm. The electronegativities of atoms H and X are χ_A and χ_B.

σ bond between the two atoms. The ionization potential for H is 1312 kJ/mol (13.6 eV), whereas that for F is 1680 kJ/mol (17.41 eV). When the wave functions for the hydrogen 1*s* and fluorine 2*p* orbital are combined, the resulting molecular orbital will have an energy that is closer to that of the fluorine orbital than to that of the hydrogen orbital. In simple terms, that means that the bonding molecular orbital is more like a fluorine orbital than a hydrogen orbital. This is loosely equivalent to saying that the electron spends more time around the fluorine atom, as we did in describing bonding in HF in valence bond terms.

The bonding in heteronuclear species can be considered as the mixing of atomic states to generate molecular orbitals with the resulting molecular orbitals having a larger contribution from the more electronegative atom. For example, the ionization potential for Li is 520 kJ/mol (5.39 eV), whereas that of hydrogen is 1312 kJ/mol (13.6 eV). Therefore, the bonding orbital in the LiH molecule will have a great deal more of the character of the hydrogen 1 σ orbital. In fact, the compound LiH is substantially ionic and we normally consider the hydrides of the group IA metals to be ionic. When we consider the compound LiF, the ionization potentials of the two atoms (energy of the atomic states for which the wave functions are being combined) are so different that the resulting "molecular orbital" is essentially the same as an atomic orbital on the fluorine atom. This means that in the compound, the electron is essentially transferred to the F atom when the bond forms. Accordingly, we consider LiF to be an ionic compound in which the species present are Li^+ and F^-.

3.6 ELECTRONEGATIVITY

As has just been described, when a covalent bond forms between two atoms, there is no reason to assume that the pair of electrons is shared *equally* between the atoms. What is needed is some sort of way to provide a relative index of the ability of an atom to attract electrons. Linus Pauling developed an approach to this problem by describing a property now known as the *electronegativity* of an atom. This property gives a measure of the tendency of an atom in a molecule to attract electrons. Pauling devised a way to give numerical values to describe this property that makes use of the fact that the covalent bonds between atoms of different electronegativity are more stable than if they were purely

covalent (with equal sharing of the electron pair). For a diatomic molecule AB, the actual bond energy, D_{AB}, is written as

$$D_{AB} = \frac{1}{2}[D_{AA} + D_{BB}] + \Delta_{AB} \tag{3.65}$$

where D_{AA} and D_{BB} are the bond energies in the purely covalent diatomic species A_2 and B_2, respectively. Because the actual bond between A and B is stronger than if the bond were purely covalent, the term Δ_{AB} corrects for the additional stability. The degree to which the sharing of the electron pair is unequal depends on the property known as electronegativity. Pauling related the additional stability of the bond to the tendency of the atoms to attract electrons by means of the equation

$$\Delta_{AB} = 96.48\,|\chi_A - \chi_B|^2 \tag{3.66}$$

In this equation, χ_A and χ_B are values that describe the electron-attracting ability (electronegativity) of atoms A and B, respectively. The constant 96.48 appears so that the value of Δ_{AB} will be given in kJ mol^{-1}. If the constant is 23.06, the value of Δ_{AB} will be in kcal mol^{-1}. Note that it is the *difference* between the values for the two atoms that is related to the additional stability of the bond. With the values of Δ_{AB} known for many types of bonds, it is possible to assign values for χ_A and χ_B, but only when there is a value known for at least one atom. Pauling solved this problem by *assigning* a value of 4.0 for the electronegativity of fluorine. In that way, the electronegativities of all other atoms are positive values between 0 and 4. Based on more recent bond energy values, the value of 3.98 is sometimes used. It would not have made any difference if the fluorine atom had been assigned a value of 100 because other atoms would then have electronegativities between 96 and 100.

With the electronegativity of the fluorine atom being assigned a value of 4.0, it was now possible to determine a value for hydrogen because the H–H and F–F bond energies were known, as was the bond energy for the H–F molecule. Using those bond energies, the electronegativity of H is found to be about 2.2. Keep in mind that it is only the *difference* in electronegativity that is related to the additional stability of the bond, not the actual values. Pauling electronegativity values for many atoms are shown in Table 3.3.

Whereas the approach just described is based on the average bond energy of A_2 and B_2 as described by the arithmetic mean, $\frac{1}{2}(D_{AA} + D_{BB})$, a different approach is based on the average bond energy being given by $(D_{AA} \times D_{BB})^{1/2}$. This is a geometric mean, which gives a value for the additional stability of the molecule as

$$\Delta' = D_{AB} - (D_{AA} \times D_{BB})^{1/2} \tag{3.67}$$

For molecules that are highly polar, this equation gives better agreement with the electronegativity difference between the atoms and the additional stability of the bond than does Eq. (3.65).

Pauling based electronegativity values on bond energies between atoms, but that is not the only way to approach the problem of the ability of atoms in a molecule to attract electrons. For example, the ease of removing an electron from an atom, the ionization potential, is related to its ability to attract electrons to itself. The electron affinity also gives a measure of the ability of an atom to hold on to an electron that it has gained. These *atomic* properties should therefore be related to the ability of an atom in a molecule to attract electrons. Therefore, it is natural to make use of these properties in an equation

Table 3.3 Pauling Electronegativities of Atoms.

H 2.2							
Li 1.0	Be 1.6		B 2.0	C 2.6	N 3.0	O 3.4	F 4.0
Na 1.0	Mg 1.3		Al 1.6	Si 1.9	P 2.2	S 2.6	Cl 3.2
K 0.8	Ca 1.0	Sc .... Zn 1.2 .... 1.7	Ga 1.8	Ge 2.0	As 2.2	Se 2.6	Br 3.0
Rb 0.8	Sr 0.9	Y .... Cd 1.1 .... 1.5	In 1.8	Sn 2.0	Sb 2.1	Te 2.1	I 2.7
Cs 0.8	Ba 0.9	La .... Hg 1.1 .... 1.5	Tl 1.4	Pb 1.6	Bi 1.7	Po 1.8	At 2.0

to express the electronegativity of an atom. Such an approach was taken by Mulliken, who proposed that the electronegativity, χ, of an atom A could be expressed as

$$\chi_A = \frac{1}{2}[I_A + E_A] \tag{3.68}$$

In this equation, I_A is the ionization potential and E_A is the electron affinity for the atom, and it is the average of these two properties that Mulliken proposes to use as the electronegativity of the atom. When the energies are expressed in electron volts, the Mulliken electronegativity for the fluorine atom is 3.91 rather than the value of 4.0 assigned by Pauling. In general, the electronegativity values on the two scales do not differ by very much.

If a property is as important as is electronegativity, it is not surprising that a large number of approaches have been taken to provide measures of the property. Although we have already described two approaches, we should also mention one additional method. Allred and Rochow made use of the equation

$$\chi_A = 0.359\left(\frac{Z^*}{r^2}\right) + 0.744 \tag{3.69}$$

In this equation, Z^* is the *effective* nuclear charge, which takes into account the fact that an outer electron is screened from experiencing the effect of the *actual* nuclear charge by the electrons that are closer to the nucleus (see Section 2.4). In principle, the Allred-Rochow electronegativity scale is based on the electrostatic interaction between valence shell electrons and the nucleus.

Probably the most important use of electronegativity values is in predicting bond polarities. For example, in the H–F bond, the shared electron pair will reside closer to the fluorine atom because it has an electronegativity of 4.0 while that of the hydrogen atom is 2.2. In other words, the electron pair is shared, but not equally. If we consider the HCl molecule, the shared electron pair will reside closer to the chlorine atom, which has an electronegativity of 3.2, but the electron pair will be shared more

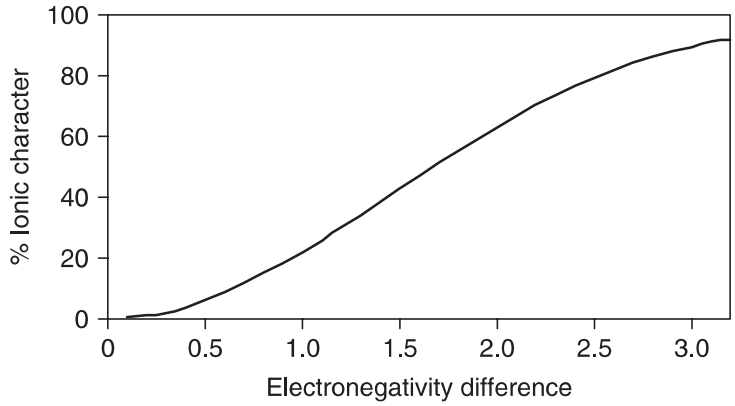


FIGURE 3.10 The variation in percent ionic character to a bond and the difference in the electronegativities of the atoms.

nearly equally than is the case for HF because the difference in electronegativity is smaller for HCl. We will have many opportunities to use this principle when describing structures of inorganic compounds.

Having shown that the weighting coefficient (λ) of the term giving the contribution of an ionic structure to the molecular wave function is related to the dipole moment of the molecule, it is logical to expect that equations could be developed that relate the ionic character of a bond to the electronegativities of the atoms. Two such equations that give the percent ionic character of the bond in terms of the electronegativities of the atoms are

$$\%\ \text{Ionic character} = 16\,|\chi_A - \chi_B| + 3.5\,|\chi_A - \chi_B|^2 \tag{3.70}$$

$$\%\ \text{Ionic character} = 18\,|\chi_A - \chi_B|^{1.4} \tag{3.71}$$

Although the equations look very different, the calculated values for the percent ionic character are approximately equal for many types of bonds. If the difference in electronegativity is 1.0, Eq. (3.70) predicts 19.5% ionic character while Eq. (3.71) gives a value of 18%. This difference is insignificant for most purposes. After one of these equations is used to estimate the percent ionic character, Eq. (3.61) can be used to determine the coefficient λ in the molecular wave function. Figure 3.10 shows how percent ionic character varies with the difference in electronegativity.

When the electrons in a covalent bond are shared equally, the length of the bond between the atoms can be approximated as the sum of the covalent radii. However, when the bond is polar, the bond is not only stronger than if it were purely covalent, it is also shorter. As shown earlier, the amount by which a polar bond between two atoms is stronger than if it were purely covalent is related to the difference in electronegativity between the two atoms. It follows that the amount by which the bond is shorter than the sum of the covalent radii should also be related to the difference in electronegativity. An equation that expresses the bond length in terms of atomic radii and the difference in electronegativity is the Schomaker-Stevenson equation. That equation can be written as

$$r_{AB} = r_A + r_B - 9.0\,|\chi_A - \chi_B| \tag{3.72}$$

where χ_A and χ_B are the electronegativities of atoms A and B, respectively, and r_A and r_B are their covalent radii expressed in picometers. This equation provides a good approximation to bond lengths. When the correction for the difference in electronegativity is applied to polar molecules, the calculated bond lengths agree considerably better with experimental values.

In this chapter, the basic ideas related to the molecular orbital approach to covalent bonds have been presented. Other applications of the molecular orbital method will be discussed in Chapters 5 and 17.

3.7 SPECTROSCOPIC STATES FOR MOLECULES

For diatomic molecules, there is coupling of spin and orbital angular momenta by a coupling scheme that is similar to the Russell-Saunders procedure described for atoms. When the electrons are in a specific molecular orbital, they have the same orbital angular momentum as designated by the m_l value. As in the case of atoms, the m_l value depends on the type of orbital. When the internuclear axis is the z-axis, the orbitals that form σ bonds (which are symmetric around the internuclear axis) are the s, p_z, and d_{z^2} orbitals. Those which form π bonds are the p_x, p_y, d_{xz}, and d_{yz} orbitals. The $d_{x^2-y^2}$ and d_{xy} can overlap in a "sideways" fashion with one stacked above the other, and the bond would be a δ bond. For these types of molecular orbitals, the corresponding m_l values are

$$\begin{aligned} \sigma: &\quad m_l = 0 \\ \pi: &\quad m_l = \pm 1 \\ \delta: &\quad m_l = \pm 2 \end{aligned}$$

As in the case of atoms, the molecular term symbol is written as ${}^{2S+1}L$, where L is the absolute value of M_L (the highest positive value). The molecular states are designated as for atoms except for the use of capital Greek letters:

$$\begin{aligned} M_L &= 0 \text{ the spectroscopic state is } \Sigma \\ M_L &= 1 \text{ the spectroscopic state is } \Pi \\ M_L &= 2 \text{ the spectroscopic state is } \Delta \end{aligned}$$

After writing the molecular orbital configuration, the vector sums are obtained. For example, in H_2 the two bonding electrons reside in a σ orbital, and they are paired so $S = +\frac{1}{2} + (-\frac{1}{2}) = 0$. As shown earlier, for a σ orbital the m_l is 0 so the two electrons combined have $M_L = 0$. Therefore, the ground state for the H_2 molecule is ${}^1\Sigma$. As in the case of atoms, all filled shells have $\Sigma s_i = 0$, which results in a ${}^1\Sigma$ state.

The N_2 molecule has the configuration $(\sigma)^2\ (\sigma^*)^2\ (\sigma)^2\ (\pi)^2\ (\pi)^2$, so all of the populated orbitals are filled. Therefore, the spectroscopic state is ${}^1\Sigma$. For O_2 the unfilled orbitals are $(\pi_x^*)^1\ (\pi_y^*)^1$ and the filled orbitals do not determine the spectroscopic state. For a π orbital, $m_l = \pm 1$. These vectors could be combined with spin vectors of $\pm\frac{1}{2}$. If both spins have the same sign, $|S| = 1$ and the state will be a triplet. If the spins are opposite, $|S| = 0$ and the state is a singlet. Because $M_l = \Sigma m_l$ and the m_l values

for the π orbitals are ± 1, the possible values for M_L are 2, 0, and -2. Possible ways to combine M_S and M_L are shown below when the values are (m_l,s):

M_L	M_S		
	1	0	-1
2		$(1,\frac{1}{2}),(1,-\frac{1}{2})$	
0	$(1,\frac{1}{2}),(-1,\frac{1}{2})$	$(1,\frac{1}{2}),(-1,-\frac{1}{2})$ $(1,-\frac{1}{2}),(-1,\frac{1}{2})$	$(1,-\frac{1}{2}),(-1,-\frac{1}{2})$
-2		$(-1,\frac{1}{2}),(-1,-\frac{1}{2})$	

Those cases for $M_L = 2$ result when the spins are *opposed* and, therefore, represent a $^1\Delta$ state. There is one combination where $M_L = 0$ with the S vector having values of $+1$, 0, and -1, which corresponds to $^3\Sigma$. The remaining combinations correspond to the $^1\Sigma$ term. Of these states ($^1\Delta$,$^1\Sigma$, and $^3\Sigma$), the one having the highest multiplicity lies lowest in energy, so the ground state of the O_2 molecule is $^3\Sigma$. The *ground state* could be identified quickly by simply placing the electrons in separate π orbitals with parallel spins and obtaining the M_L and M_S values.

For the CN molecule, the configuration is $(\sigma)^2\ (\sigma_z)^2\ (\pi_x)^2\ (\pi_y)^2\ (\sigma_z)^1$. The single electron in the σ_z orbital gives $M_L = 0$ and $S = \frac{1}{2}$, so the ground state is $^3\Sigma$. Several species such as N_2, CO, NO^+, and CN^- have the configuration $(\sigma)^2\ (\sigma_z)^2\ (\pi_x)^2\ (\pi_y)^2\ (\sigma_z)^2$, which is a closed-shell arrangement. Therefore, the ground state for these species is $^1\Sigma$. The NO molecule has the configuration $(\sigma)^2\ (\sigma_z)^2\ (\pi_x)^2\ (\pi_y)^2\ (\sigma_z)^2(\pi_x{}^*)^1$, which gives rise to $S = \frac{1}{2}$ and $M_L = 1$. These values give rise to a ground state that is $^2\Pi$.

REFERENCES FOR FURTHER STUDY

Cotton, F. A., Wilkinson, G., and Murillo, C. A. (1999). *Advanced Inorganic Chemistry*, 6th ed. John Wiley, New York. Almost 1400 pages devoted to all phases of inorganic chemistry. An excellent reference text.

DeKock, R. L., and Gray, H. B. (1980). *Chemical Bonding and Structure*. Benjamin Cummings, Menlo Park, CA. One of the best introductions to bonding available. Highly recommended.

Greenwood, N. N., and Earnshaw, A. (1997). *Chemistry of the Elements*, 2nd ed. Butterworth-Heinemann, New York. Although this is a standard reference text on descriptive chemistry, it contains an enormous body of information on bonding.

House, J. E. (2003). *Fundamentals of Quantum Chemistry*. Elsevier, New York. An introduction to quantum-mechanical methods at an elementary level that includes mathematical details.

Lide, D. R., Ed. (2003). *CRC Handbook of Chemistry and Physics*, 84th ed. CRC Press, Boca Raton, FL. Various sections in this massive handbook contain a large amount of data on molecular parameters.

Lowe, J. P. (1993). *Quantum Chemistry*, 2nd ed. Academic Press, New York. This is an excellent book for studying molecular orbital methods at a higher level.

Mackay, K., Mackay, R. A., and Henderson, W. (2002). *Introduction to Modern Inorganic Chemistry*, 6th ed. Nelson Thornes, Cheltenham, UK. One of the very successful standard texts in inorganic chemistry.

Mulliken, R. S., Rieke, A., Orloff, D., and Orloff, H. (1949). Overlap integrals and chemical binding. *J. Chem. Phys.* **17**, 510, and Formulas and numerical tables for overlap integrals, *J. Chem. Phys.* **17**, 1248–1267. These two papers present the basis for calculating overlap integrals and show the extensive tables of calculated values.

Pauling, L. (1960). *The Nature of the Chemical Bond*, 3rd ed. Cornell University Press, Ithaca, NY. Although somewhat dated in some areas, this is a true classic in bonding theory.

Sharpe, A. G. (1992). *Inorganic Chemistry*, 3rd ed. Longman, New York. Excellent coverage of bonding concepts in inorganic molecules.

QUESTIONS AND PROBLEMS

1. For each of the following, draw a molecular orbital energy level diagram and give the bond order. Tell whether the species would be more or less stable after gaining an electron. (a) O_2^+; (b) CN; (c) S_2; (d) NO; (e) Be_2^+.
2. Explain in terms of molecular orbitals why Li_2 is stable but Be_2 is not.
3. Which has the greater bond energy, NO or C_2? Explain by making appropriate drawings.
4. Numerical data are given below for the BN and BO molecules. Match the properties to these molecules and explain your answers.

   Data: 120 pm, 128 pm, 8.0 eV, 4.0 eV.
5. If the H–H and S–S bond energies are 266 and 432 kJ mol^{-1}, respectively, what would be the H–S bond energy?
6. The stretching vibration for NO is found at 1876 cm^{-1}, whereas that for NO^+ is at 2300 cm^{-1}. Explain this difference.
7. What is the Cl–F bond length if the covalent radii of Cl and F are 99 and 71 pm, respectively? Explain your answer in terms of resonance.
8. Consider a diatomic molecule A_2 in which there is a single σ bond. Excitation of an electron to the σ^* state gives rise to an absorption at 15,000 cm^{-1}. The binding energy of an electron in the valence shell of atom A is −9.5 eV.
   (a) If the overlap integral has a value of 0.12, determine the value of the exchange integral, H_{12}.
   (b) Calculate the actual values of the bonding and antibonding molecular orbitals for the A_2 molecule.
   (c) What is the single bond energy in the A_2 molecule?
9. Arrange the species O_2^{2-}, O_2^+, O_2, and O_2^- in the order of decreasing bond length. Explain this order in terms of molecular orbital populations.
10. Explain why the electron affinity of the NO molecule is 88 kJ mol^{-1} but that of the CN molecule is 368 kJ mol^{-1}.
11. In the spectrum of the CN molecule, an absorption band centered around 9,000 cm^{-1} appears. Explain the possible origin of this band in terms of the molecular orbitals in this molecule. What type of transition is involved?
12. Consider the Li_2 molecule, which has a dissociation energy of 1.03 eV. The first ionization potential for the Li atom is 5.30 eV. Describe the bonding in Li_2 in terms of a molecular orbital energy diagram. If a value of 0.12 is appropriate for the overlap integral, what is the value of the exchange integral?
13. For a molecule XY, the molecular wave function can be written as

$$\psi_{\text{molecule}} = \psi_{\text{covalent}} + 0.70\ \psi_{\text{ionic}}$$

   Calculate the percent of ionic character in the X–Y bond. If the X–Y bond length is 142 pm, what is the dipole moment of XY?

14. What Pauling electronegativity is predicted for an element X if the H–X bond energy is 402 kJ mol^{-1}? The H–H bond energy is 432 kJ mol^{-1} and the X–X bond energy is 335 kJ mol^{-1}. What would be the percent ionic character of the H–X bond? If the molecular wave function is written as

$$\psi_{\text{molecule}} = \psi_{\text{covalent}} + \lambda\, \psi_{\text{ionic}}$$

what is the value of λ?

15. Suppose the bond energies of A_2 and X_2 are 210 and 345 kJ mol^{-1}, respectively. If the electronegativities of A and X are 2.0 and 3.1, respectively, what will be the strength of the A–X bond? What will be the dipole moment if the internuclear distance is 125 pm?

16. For a molecule XY, the molecular wave function can be written as

$$\psi_{\text{molecule}} = \psi_{\text{covalent}} + 0.50\, \psi_{\text{ionic}}$$

Calculate the percent ionic character of the X–Y bond. If the bond length is 148 pm, what is the dipole moment of XY?

17. Determine the spectroscopic ground states for the following diatomic molecules: (a) BN; (b) C_2^+; (c) LiH; (d) CN^-; (e) C_2^-.